

DR-2 for the lattice class – a decoupling-free closure of the sphere

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1. Purpose and scope

The two general-position routes to DR-2 (decoupling, dr2-cross-scale-induction v1.1; and the elementary circle-richness reduction, dr2-circle-richness-reduction v1.0) leave the arbitrary- Q problem OPEN: the former cites the hard core of Bourgain–Demeter, and the latter reduces DR-2 to bounding the sum-circle richness $T'(Q)$, which can be as large as N for adversarial real configurations. This note observes that the configurations DR-2 must handle IN TECT are not adversarial: they are lattice points on a sphere (BCC/FCC reciprocal-lattice momentum shells), and for these $T'(Q)$ is controlled UNCONDITIONALLY by a classical divisor bound. The result is Route A, self-contained and decoupling-free.

2. The lattice reduction

Let $\Lambda \subset \mathbb{R}^3$ be a fixed lattice (take $\Lambda = \mathbb{Z}^3$; the argument is lattice-independent) and

$$Q = \Lambda \cap \{x : |x|^2 = R\}, \quad N = |Q|.$$

Every sum $m = a + b$ with $a, b \in Q$ lies in $\Lambda + \Lambda = \Lambda$ (an integer vector), and the sum-level circle of Lemma A is

$$C_m = \{x : |x|^2 = R, \ x \cdot m = \tfrac{1}{2}|m|^2\}.$$

For m with $r(m) > 0$, $\frac{1}{2}|m|^2 = R + a \cdot b \in \mathbb{Z}$, so C_m is the intersection of the sphere with a RATIONAL plane: a circle whose Λ -points form a coset of the rank-2 sublattice $\Lambda \cap m^\perp$. Writing $|x|^2$ in a basis of that sublattice expresses $\#(Q \cap C_m)$ as the number of representations of a fixed integer by a positive-definite BINARY QUADRATIC FORM of discriminant $O(R)$.

[DIV-CIRC] (classical). The number of lattice points on a circle of squared radius $\leq R$ – equivalently the number of representations of an integer $n \leq R$ by a positive-definite binary quadratic form of discriminant $O(R)$ – is $O_\epsilon(R^\epsilon)$, uniformly. (For $\Lambda = \mathbb{Z}^2$ this is the divisor bound $r_2(n) \ll_\epsilon n^\epsilon$, provable from unique factorisation in $\mathbb{Z}[i]$; the binary-form version follows from the divisor bound for ideal norms in the associated order.)

3. Theorem (lattice DR-2, conditional on [DIV-CIRC])

Theorem. For $Q = \Lambda \cap \{|x|^2 = R\}$, conditional on [DIV-CIRC],

$$E_+(Q) \leq (1 + C_\epsilon R^\epsilon) N^2.$$

In particular, for Gauss-typical shells $R \asymp N^{2+o(1)}$ (the regime $N \gg R^\delta$ holds), $E_+(Q) \lesssim_\epsilon N^{2+\epsilon}$ – DR-2 holds for the lattice class, with NO decoupling.

Proof. By Lemma A (R-025), $E_+(Q) \leq (1 + T'(Q))N^2$ with $T'(Q) = \max_{m \neq 0} \#(Q \cap C_m)$. Each C_m is a circle on the sphere of squared radius R , so by §2 and [DIV-CIRC], $\#(Q \cap C_m) \leq \#(\Lambda \cap C_m) \leq C_\epsilon R^\epsilon$ uniformly in m . Hence $T'(Q) \leq C_\epsilon R^\epsilon$ and the bound follows. For $R \asymp N^2$, $R^\epsilon \asymp N^{2\epsilon}$, giving $E_+(Q) \lesssim_\epsilon N^{2+\epsilon}$. $\square$

4. Numerical verification (exact arithmetic)

dr2_lattice_divisor.py (5/5) computes E_+ and T' EXACTLY (integers, no tolerance) on $\mathbb{Z}^3$ shells $R = 101, \dots, 9974$ ($N = 168, \dots, 2040$):

R	N	E_+/N^2	T'	T'/N
101	168	5.27	18	0.107
909	504	4.93	24	0.048
1826	672	4.10	24	0.036
4994	1152	3.93	32	0.028
9974	2040	3.92	48	0.024

The lemma assert $E_+ \leq (1 + T')N^2$ holds on every shell (the proof check on real arithmetic data). T' grows sub-linearly: T'/N falls monotonically $0.107 \rightarrow 0.024$ and the log-log slope is 0.177 with the per-point exponent $\log T' / \log R$ falling $0.63 \rightarrow 0.42$ – the signature of $R^{o(1)}$. E_+/N^2 stays ≤ 5.3 (energy is $O(N^2)$, the R^ϵ ceiling is never approached). $\mathbb{Z}^3$ and the FCC sublattice give identical (N, T', E_+) at $R = 1826$: the mechanism is lattice-independent, so it applies to the BCC/FCC momentum shells of TECT.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

Four concrete objections, found before submission:

- (α) “Is [DIV-CIRC] really uniform over the varying rational planes?” RECORDED as the explicit named hypothesis. The $\mathbb{Z}^2$ case is textbook; the binary-form version is uniform because the forms have discriminant $O(R)$ and the representation count is $\ll_\epsilon (n \cdot \text{disc})^\epsilon \ll_\epsilon R^\epsilon$. The grade is T6 CONDITIONAL on [DIV-CIRC] precisely so this classical input is named, not hidden; the operator may elevate to T7.
- (β) “The bound is $R^\epsilon N^2$, which is $N^{2+\epsilon}$ only when $R \lesssim N^{O(1)}$.” VALID and scoped: for very empty shells ($N \ll R^{1/2}$) the R^ϵ factor is not absorbed, but there the absolute energy is small anyway; the DR-2-relevant (Gauss-typical, dominant) shells satisfy $R \asymp N^{2+o(1)}$ and are covered. The theorem is stated as $E_+ \leq (1 + C_\epsilon R^\epsilon)N^2$ to avoid overclaiming the $N^{2+\epsilon}$ form universally.
- (γ) “Does this close DR2-SHARE / discharge H-ADM-COH?” NOT asserted. This closes the ADDITIVE-ENERGY form of DR-2 for the lattice class. The STEP-5B carrier richness $\chi(P)$ and T' are both rich-circle quantities and are closely related, but their exact equivalence is not established here, so whether the lattice closure discharges H-ADM-COH is an OPERATOR integration decision. NO gate/tier/hypothesis flip is made.

(δ) “Arbitrary- Q DR-2?” STILL OPEN. This note covers the lattice class only; the rich-latitude example ($T'=N$) shows general Q is not reached by this route. The decoupling route (T4+) remains the general-case path.

Quantitative sanity check (mandatory). Exact integer arithmetic (no tolerance): the lemma assert $E_+ \leq (1 + T')N^2$ holds on all six shells. Magnitude: $E_+/N^2 \in [3.9, 5.3]$ (bounded, $\ll N$). Growth: $T'/N \rightarrow 0$ ($0.107 \rightarrow 0.024$), log-log slope $0.177 < 1$ (sub-polynomial). Limit/consistency: $\mathbb{Z}^3 = \text{FCC}$ at $R = 1826$ ($N = 672, T' = 24$). Direction: $T' \geq 1$, all terms positive, upper bound.

6. Result footer

Result ID:	R-026 / DR-2 for the lattice class (Route A)
Precise statement:	For $Q = \text{Lambda cap } \{ x ^2=R\}$ (Lambda a fixed lattice; $\mathbb{Z}^3$ / BCC / FCC), conditional on [DIV-CIRC] (lattice points on a circle of sq-radius $\leq R$ number $O_\epsilon(R^\epsilon)$, classical), $E_+(Q) \leq (1 + C_\epsilon R^\epsilon) N^2$ by Lemma A (R-025). For Gauss-typical shells $R \sim N^2$ this is $E_+(Q) \leq_\epsilon N^{2+\epsilon}$ -- DR-2, DECOUPLING-FREE.
Scope:	The TECT DR-2 carrier (BCC/FCC momentum-shell lattice points); additive-energy form of DR-2.
Dependencies:	Lemma A (R-025, unconditional) + [DIV-CIRC] (classical divisor bound for lattice points on circles). NO decoupling, NO PSM conjecture.
Evidence grade:	T6 PROVED CONDITIONAL on [DIV-CIRC] (textbook number theory; operator may elevate to T7). Verified exactly (dr2_lattice_divisor.py 5/5).
Reproduction command:	python codes/vacuum/dr2_lattice_divisor.py
Expected output:	5/5 PASS; $E_+ \leq (1+T')N^2$ on all shells; $T'/N \rightarrow 0$ ($0.107 \rightarrow 0.024$); log-log slope 0.177; $E_+/N^2 \leq 5.3$.
Falsification gate:	A lattice shell with $E_+(Q) > (1+T'(Q))N^2$ (refutes Lemma A) or T' growing like N^δ , $\delta > 0$ (refutes [DIV-CIRC]); neither observed.
Tier before / after:	NEW result for the lattice class: T6 CONDITIONAL on [DIV-CIRC]. General DR-2 unchanged (T4+). NO flip: DR2-SHARE OPEN, B5 T5, B1 T6, H-ADM-COH retained.
No-overclaim statement:	Closes the additive-energy DR-2 for the LATTICE class only, conditional on a classical bound. Does NOT close arbitrary- Q DR-2 and does NOT by itself discharge H-ADM-COH; the T' -to- $\chi(P)$ equivalence and any gate/hypothesis flip are operator decisions.
Next required action:	Operator decision: (i) accept [DIV-CIRC] ($\rightarrow$ T7) and (ii) determine whether lattice-class closure + the carrier-richness equivalence discharges H-ADM-COH in B1.